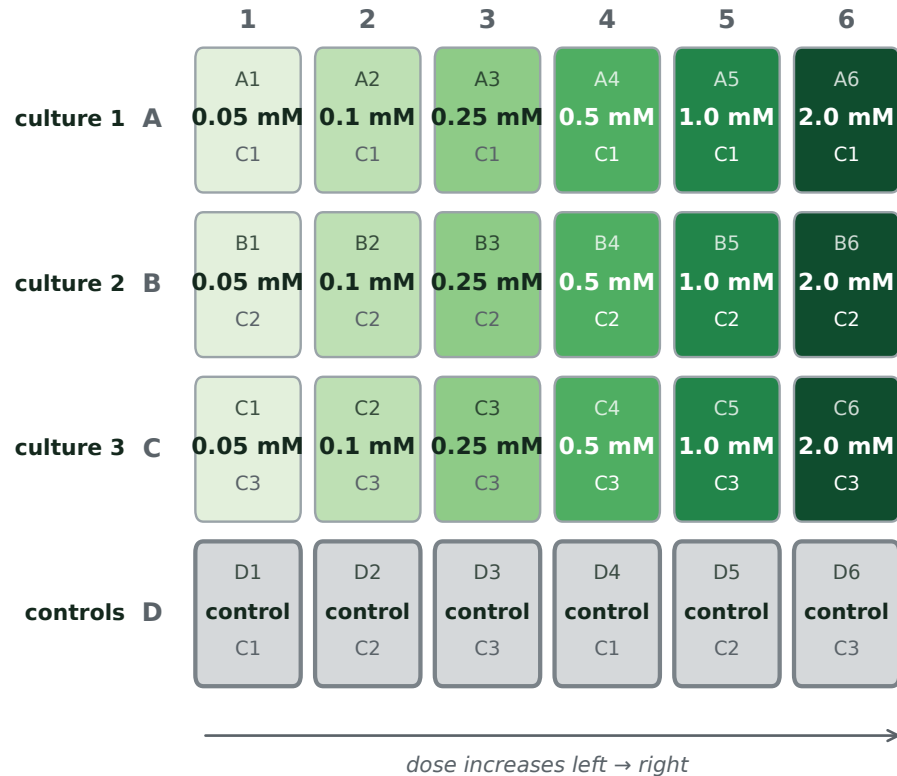


Plates 1 and 2 — n-propyl gallate dose series

one row per culture • one dose per column • bottom row all controls • both plates filled identically

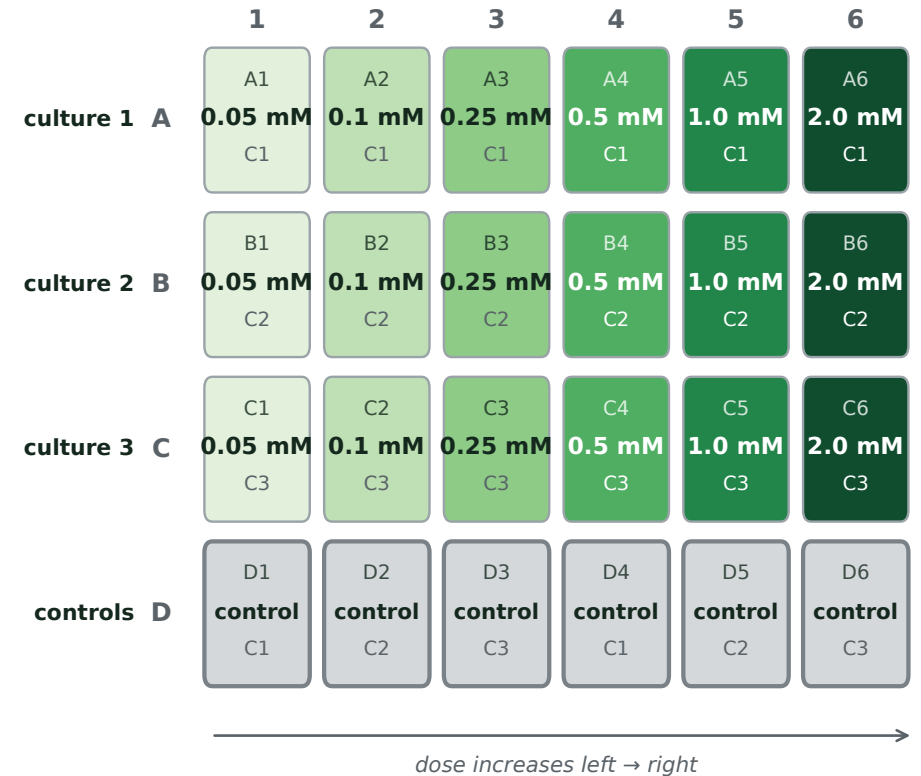
15 °C

fill left to right, row by row



27 °C

same layout as the 15 °C plate



Legend



Top dose is 20x the maximum in the first nPG run. Range spans 1.6 decades against 0.90 before, so the slope has a longer lever arm.

Dose is tied to plate column. Row D is the diagnostic: if D1 to D6 trend across the row, the plate has a gradient and the dose axis has inherited it.

Plate 3 — azide-resistant respiration

27 °C only • 10 no-cell / 7 azide / 7 azide + SHAM • interleaved so every column carries all three

27 °C

fill in three passes: all B, then all A, then all AS

	1	2	3	4	5	6
A	A1 az + SHAM 1 + 0.2	A2 no cells medium	A3 az + SHAM 1 + 0.2	A4 no cells medium	A5 no cells medium	A6 azide 1 mM
B	B1 azide 1 mM	B2 azide 1 mM	B3 no cells medium	B4 azide 1 mM	B5 az + SHAM 1 + 0.2	B6 no cells medium
C	C1 no cells medium	C2 az + SHAM 1 + 0.2	C3 no cells medium	C4 azide 1 mM	C5 azide 1 mM	C6 az + SHAM 1 + 0.2
D	D1 az + SHAM 1 + 0.2	D2 no cells medium	D3 azide 1 mM	D4 az + SHAM 1 + 0.2	D5 no cells medium	D6 no cells medium

The two comparisons

1. 7 azide wells vs 10 no-cell wells
Does anything respire once cytochrome oxidase is blocked?
2. 7 azide vs 7 azide + SHAM
Is that residual SHAM-sensitive? Azide-resistant AND SHAM-sensitive is the operational definition of alternative oxidase activity.

Why this design and not the last one

Last run had 3 usable blanks reading 0.0095, 0.0190 and 0.0280 mg/L/h. The residual was 0.0093, so the baseline uncertainty was the same size as the signal and the comparison gave $P = 0.22$.

Ten blanks cut that uncertainty roughly in half. The same residual then reaches $t = 3.0$, $P = 0.013$, with power around 0.78 at the density you can use.

Blanks get the extra wells rather than the treatment groups because the baseline is where all the uncertainty sits: blank sd 0.0093 against azide sd 0.0021.

Interleaving controls position by layout. Positions C4 and C5 read low on both A plates last time regardless of contents.

Azide at 1 mM, not 5

Your titration gave identical rates from 0.5 to 10 mM, so the block saturates at the bottom of that range. 1 mM is the standard concentration and avoids the general toxicity risk at 5 to 10 mM.

	no cells (plain medium) n = 10		azide 1 mM n = 7		azide 1 mM + SHAM 0.2 mM n = 7
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Every vial on this plate: culture (or plain medium for B) + 50 µL aqueous slot + 20 µL ethanol slot.

B gets 50 µL water + 20 µL ethanol. A gets 50 µL azide 100 mM + 20 µL ethanol. AS gets 50 µL azide 100 mM + 20 µL SHAM 50 mM.

Solvent identical in all 24: 0.39% ethanol, 0.99% water.

nPG plates — well by well (same for 15 °C and 27 °C)

Every vial: your usual culture volume (~5 mL) + 20 µL from the stock column. Solvent identical in all 24.

well	culture	20 µL slot	final	note
 A1	culture 1	nPG stock 12.5 mM	0.05 mM	in first run too
 A2	culture 1	nPG stock 25 mM	0.1 mM	in first run too
 A3	culture 1	nPG stock 62.5 mM	0.25 mM	NEW dose
 A4	culture 1	nPG stock 125 mM	0.5 mM	NEW dose
 A5	culture 1	nPG stock 250 mM	1.0 mM	NEW dose
 A6	culture 1	nPG stock 500 mM	2.0 mM	NEW dose
 B1	culture 2	nPG stock 12.5 mM	0.05 mM	in first run too
 B2	culture 2	nPG stock 25 mM	0.1 mM	in first run too
 B3	culture 2	nPG stock 62.5 mM	0.25 mM	NEW dose
 B4	culture 2	nPG stock 125 mM	0.5 mM	NEW dose
 B5	culture 2	nPG stock 250 mM	1.0 mM	NEW dose
 B6	culture 2	nPG stock 500 mM	2.0 mM	NEW dose
 C1	culture 3	nPG stock 12.5 mM	0.05 mM	in first run too
 C2	culture 3	nPG stock 25 mM	0.1 mM	in first run too
 C3	culture 3	nPG stock 62.5 mM	0.25 mM	NEW dose
 C4	culture 3	nPG stock 125 mM	0.5 mM	NEW dose
 C5	culture 3	nPG stock 250 mM	1.0 mM	NEW dose
 C6	culture 3	nPG stock 500 mM	2.0 mM	NEW dose
 D1	culture 1	ethanol (no compound)	0 mM	
 D2	culture 2	ethanol (no compound)	0 mM	
 D3	culture 3	ethanol (no compound)	0 mM	
 D4	culture 1	ethanol (no compound)	0 mM	
 D5	culture 2	ethanol (no compound)	0 mM	
 D6	culture 3	ethanol (no compound)	0 mM	

Azide plate — well by well (27 °C)

Every vial: ~5 mL base + 50 µL aqueous slot + 20 µL ethanol slot. Solvent identical in all 24.

well	base	additions	final	note
 A1	cells	50 µL azide 100 mM + 20 µL SHAM 50 mM	azide 1 + SHAM 0.2	
 A2	PLAIN MEDIUM	50 µL water + 20 µL ethanol	—	baseline
 A3	cells	50 µL azide 100 mM + 20 µL SHAM 50 mM	azide 1 + SHAM 0.2	
 A4	PLAIN MEDIUM	50 µL water + 20 µL ethanol	—	baseline
 A5	PLAIN MEDIUM	50 µL water + 20 µL ethanol	—	baseline
 A6	cells	50 µL azide 100 mM + 20 µL ethanol	azide 1 mM	
 B1	cells	50 µL azide 100 mM + 20 µL ethanol	azide 1 mM	
 B2	cells	50 µL azide 100 mM + 20 µL ethanol	azide 1 mM	
 B3	PLAIN MEDIUM	50 µL water + 20 µL ethanol	—	baseline
 B4	cells	50 µL azide 100 mM + 20 µL ethanol	azide 1 mM	
 B5	cells	50 µL azide 100 mM + 20 µL SHAM 50 mM	azide 1 + SHAM 0.2	
 B6	PLAIN MEDIUM	50 µL water + 20 µL ethanol	—	baseline
 C1	PLAIN MEDIUM	50 µL water + 20 µL ethanol	—	baseline
 C2	cells	50 µL azide 100 mM + 20 µL SHAM 50 mM	azide 1 + SHAM 0.2	
 C3	PLAIN MEDIUM	50 µL water + 20 µL ethanol	—	baseline
 C4	cells	50 µL azide 100 mM + 20 µL ethanol	azide 1 mM	
 C5	cells	50 µL azide 100 mM + 20 µL ethanol	azide 1 mM	
 C6	cells	50 µL azide 100 mM + 20 µL SHAM 50 mM	azide 1 + SHAM 0.2	
 D1	cells	50 µL azide 100 mM + 20 µL SHAM 50 mM	azide 1 + SHAM 0.2	
 D2	PLAIN MEDIUM	50 µL water + 20 µL ethanol	—	baseline
 D3	cells	50 µL azide 100 mM + 20 µL ethanol	azide 1 mM	
 D4	cells	50 µL azide 100 mM + 20 µL SHAM 50 mM	azide 1 + SHAM 0.2	
 D5	PLAIN MEDIUM	50 µL water + 20 µL ethanol	—	baseline
 D6	PLAIN MEDIUM	50 µL water + 20 µL ethanol	—	baseline

Stocks and notes

n-propyl gallate, ethanol stocks (M = 212.20 g/mol)

500 mM	106.1 mg in 1.00 mL absolute ethanol	20 μ L $\rightarrow$ 1.99 mM ** check it dissolves **
250 mM	500 μ L of 500 mM + 500 μ L ethanol	20 μ L $\rightarrow$ 1.00 mM
125 mM	500 μ L of 250 mM + 500 μ L ethanol	20 μ L $\rightarrow$ 0.50 mM
62.5 mM	500 μ L of 125 mM + 500 μ L ethanol	20 μ L $\rightarrow$ 0.25 mM
25 mM	400 μ L of 62.5 mM + 600 μ L ethanol	20 μ L $\rightarrow$ 0.10 mM
12.5 mM	500 μ L of 25 mM + 500 μ L ethanol	20 μ L $\rightarrow$ 0.05 mM

120 μ L of each dose stock across both plates. Make 1 mL of each. Ethanol for controls: 240 μ L.

Wrap in foil, cap tight, 4 °C. Check the colour before use: colourless to faint straw is fine, yellow means remake.

Azide plate stocks

azide 100 mM	100 μ L of your 1 M azide + 900 μ L water	50 μ L $\rightarrow$ 0.99 mM (14 wells, make 1.5 mL)
SHAM 50 mM	200 μ L of 250 mM SHAM + 800 μ L ethanol	20 μ L $\rightarrow$ 0.20 mM (7 wells, make 500 μ L)
water	for the 10 no-cell wells	50 μ L each, 500 μ L total
ethanol	for the 17 non-SHAM wells	20 μ L each, 340 μ L total

At the bench

- C1, C2, C3 on the nPG plates must be three separately grown cultures, not one suspension split three ways.
- The azide plate can come from a single suspension at OD 0.005. It asks a technical question, not a biological one.
- Fill the azide plate in three passes, all B wells then all A then all AS, so the interleaving is not a burden.
- Record which physical plate went to which incubator, and the start time for each plate separately.
- Run 24 h. Do not stop early: the azide wells accumulate signal slowly and need the full window.

When the data comes in

1. nPG plates: read row D first. Six controls across all six columns. A trend across that row means the plate has a left-right gradient and the dose axis has inherited it, because dose is tied to column.
2. Fit the windows by rule, not by eye. scripts/42_uniform_window_analysis.R applies seven rules to every curve identically. Choose one as the primary analysis before looking at the result.
3. Azide plate: compare the 7 azide wells against the 10 no-cell wells first. If they do not separate, there is no azide-resistant respiration and the SHAM comparison is moot.

What the azide plate cannot tell you

There are no nPG wells on it, so it says nothing about propyl gallate. And if the azide wells do separate from the blanks, that residual is azide-resistant respiration by definition but is only alternative oxidase if SHAM removes it. Both halves are needed.